

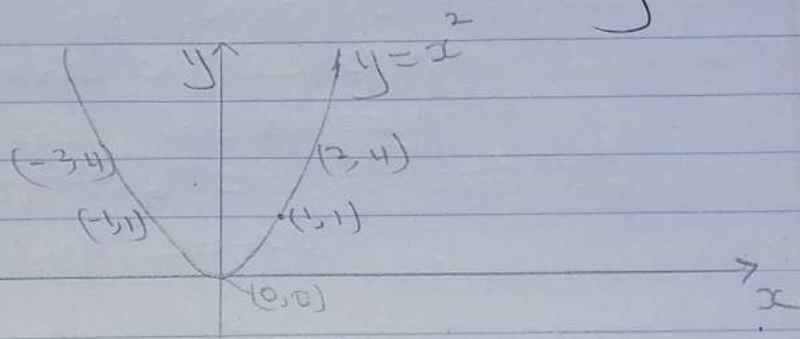
Test for Symmetry:

(103)

The following tests are useful for investigating the three types of Symmetry for graphs of equations in x and y .

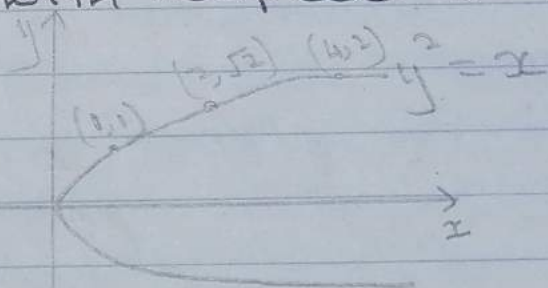
1. The graph of an equation is Symmetric with respect to the y -axis if Substitution of $-x$ for x leads to an equivalent equation

e.g. Consider the graph of $y = x^2$.
If x is substituted with $-x$
 $\Rightarrow y = (-x)^2 = x^2 \Rightarrow$ the graph is Symmetric with respect to y -axis



2. The graph of an equation is Symmetric with respect to the x -axis if substitution of $-y$ for y leads to an equivalent equation.

e.g. Consider the graph of $y^2 = x$.
If $y = -y \Rightarrow (-y)^2 = x \Rightarrow y^2 = x$.
 $\Rightarrow$ Symmetric with respect to x -axis



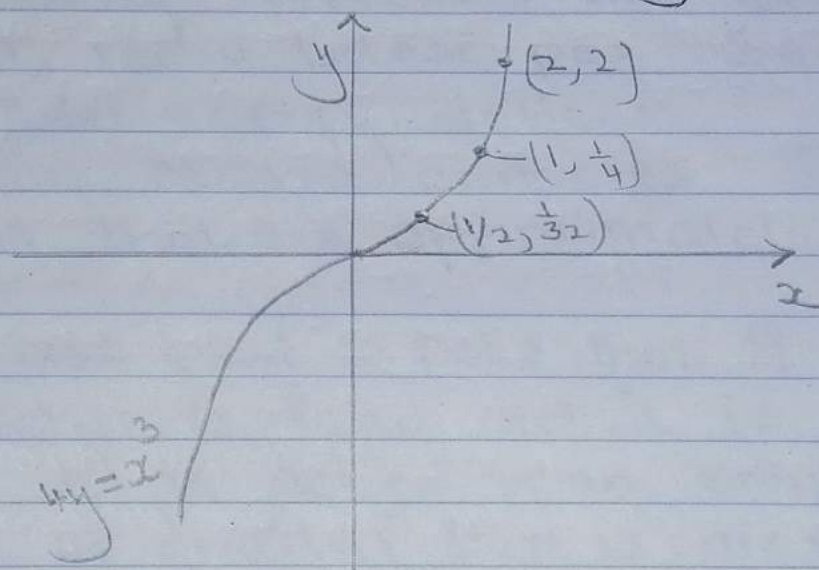
(104)

3. The graph of an equation is symmetric with respect to the origin if the simultaneous substitution of $-x$ for x and $-y$ for y leads to an equivalent equation.

e.g. consider the function $4y = x^3$.

If we substitute $-x$ for x and $-y$ for y , then

$$4(-y) = (-x)^3 \quad \text{or} \quad -4y = -x^3 \\ = 4y = x^3.$$



Conic Sections

The circle, parabola, ellipse, and hyperbola are curves that have played a special role in Mathematics.

These four curves are known as conic sections or simply conics.

The conic sections are curves resulting from planes passing through right circular cones.

(105)

Present-day application of conic sections are found in such diverse fields as art, architecture, astronomy, engineering, and physics.

They play an important role in atomic and electromagnetic field theory, as well as optics.

The orbits of planets around the sun and satellites around the earth are elliptical.

Parabolic reflectors are used for radar, radio telescopes, searchlights, and solar energy devices.

Certain atomic particles follow paths that are approximately hyperbolic.

It is also good to note that if a projectile is fired and it is assumed that it is acted upon only by the force of gravity (that is, air resistance and other outside factors are ignored), then the path of the projectile is parabolic.

Circle:

(106)

A Circle is the set of all points at a fixed distance (the radius) from a fixed point (the center).

Suppose the radius is r and the center is (h, k) .

Then a point $P(x, y)$ lies on the circle if and only if the distance from (h, k) to (x, y) is r .

$$\text{i.e. } \sqrt{(x-h)^2 + (y-k)^2} = r$$

$$\text{or } (x-h)^2 + (y-k)^2 = r^2 \dots \dots \dots (*)$$

The above is called the standard form of the equation of a circle.

If the center is at the origin, the equation assumes the particularly simple form

$$x^2 + y^2 = r^2.$$

If $r=1 \Rightarrow x^2 + y^2 = 1$ called the unit circle.

Expanding the (*) we obtain an equation of the form

$$x^2 + y^2 + ax + by + c = 0.$$

Where a, b, c are real numbers.

Example 1.

(107)

Find an equation of the circle with center $(-2, 3)$ which passes through the point $(4, 5)$.

Solution

Since the point $(4, 5)$ is on the circle, then the radius r is $d(c, d)$.

$$\text{i.e. } r = d(c, d) = \sqrt{(-2-4)^2 + (3-5)^2}$$

$$\Rightarrow r = \sqrt{40} \Rightarrow r^2 = 40$$

Using (x) with $h = -2$ and $k = 3$, we obtain

$$(x+2)^2 + (y-3)^2 = 40$$

$$x^2 + 4x + 4 + y^2 - 6y + 9 = 40$$

$$x^2 + y^2 + 4x - 6y - 27 = 0.$$

Notice that the above equation is of the form

$$x^2 + y^2 + ax + by + c = 0.$$

Conversely, if we begin with such an equation, it is always possible, by completing the squares in x and y , to obtain an equation of the form

$$(x-h)^2 + (y-k)^2 = d.$$

(108)
If $d > 0$, the graph is a circle with center (h, k) and radius $r = \sqrt{d}$.

If $d = 0$, then, since $(x-h)^2 \geq 0$ and $(y-k)^2 \geq 0$, the only solution of the equation is (h, k) , and hence the graph consists of only one point.

- Finally, if $d < 0$, the equation has no real solutions and hence there is no graph.

Example 2:

Find the center and radius of the circle with equation

$$x^2 + y^2 - 4x + 6y - 3 = 0.$$

Solution

- By rearranging the equation, we obtain

$$(x^2 - 4x) + (y^2 + 6y) = 3.$$

Next we complete the squares in x and in y .

To obtain an equivalent equation we must add the numbers to both sides of the equation.

In order to complete the square for an expression of the form $x^2 + ax$,

(109)

We add the square of half the coefficient of x , that is, add $\left(\frac{a}{2}\right)^2$, to both sides of the equation.

Similarly, for $y^2 + by$, we add $\left(\frac{b}{2}\right)^2$ to both sides.

$$\text{i.e. } (x^2 - 4x + 4) + (y^2 + 6y + 9) = 3 + 4 + 9$$

$$(x-2)^2 + (y+3)^2 = 16$$

$$(x-2)^2 + (y+3)^2 = 4^2$$

$\Rightarrow$ Equation of a circle with center as $(2, -3)$ and the radius 4.

Example 3

Determine the nature of the graph of the equation

$$x^2 + y^2 - 2x + 4y - 4 = 0.$$

Solution

$$x^2 + y^2 - 2x + 4y = 4$$

$$x^2 - 2x + y^2 + 4y = 4$$

$$(x^2 - 2x + 1) + (y^2 + 4y + 4) = 4 + 1 + 4$$

$$(x-1)^2 + (y+2)^2 = 9$$

$\Rightarrow$ Equation of a circle with $C(1, -2)$, $r=3$.

Example 4:

(110)

Find the center and radius of the circle with the given equation

$$9x^2 + 9y^2 - 6x + 12y - 31 = 0$$

Solution

$$9x^2 - 6x + 9y^2 + 12y = 31$$

Dividing all through with 9

$$x^2 - \frac{2}{3}x + y^2 + \frac{4}{3}y = \frac{31}{9}$$

$$\left(x^2 - \frac{2}{3}x + \frac{1}{9}\right) + \left(y^2 + \frac{4}{3}y + \frac{4}{9}\right) = \frac{31}{9} + \frac{1}{9} + \frac{4}{9}$$

$$\left(x - \frac{1}{3}\right)^2 + \left(y + \frac{2}{3}\right)^2 = 2^2$$

$\Rightarrow$ Equation of a Circle with Center $\left(\frac{1}{3}, -\frac{2}{3}\right)$ and radius $r=2$.

Exercise:

1. Find an equation of a circle satisfying the stated conditions

(a) Center $C(3, -2)$, radius 4.

(b) Endpoints of a diameter $A(4, -3)$ and $B(-2, 7)$

(111)

(2) Find the center and radius of the circle with the given equation.

(a) $x^2 + y^2 - 10x + 2y + 22 = 0$

(b) $2x^2 + 2y^2 - x + y - 3 = 0$.

(3) Find the equation of the tangent line to the circle $x^2 + y^2 - 6x + 8y + 17 = 0$ at the point $(1, -2)$.

(4) Prove that the line $3x - 4y + 15 = 0$ is tangent to the circle $x^2 + y^2 - 4x + 2y - 20 = 0$ at the point $(-1, 3)$.

Parabola

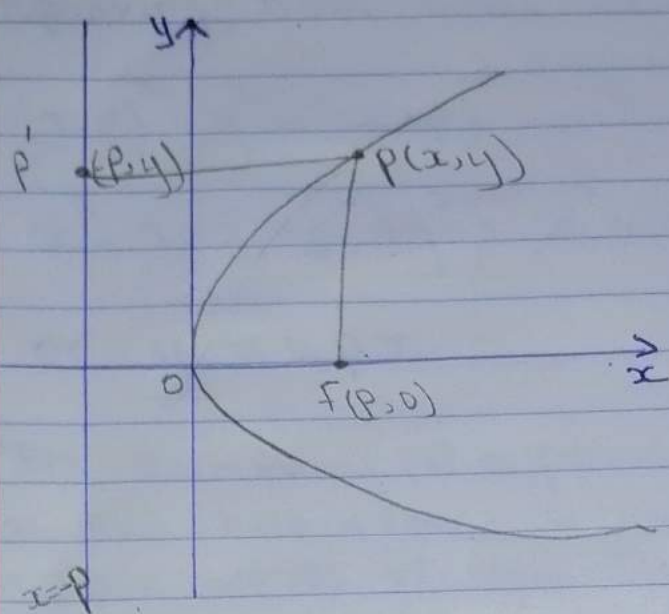
A parabola is the set of all points in a plane equidistant from a fixed line and a fixed point not on the line.

The fixed line is called the directrix of the parabola, and the fixed point is called the focus.

The line through the focus perpendicular to the directrix is called the axis, and the point where the parabola crosses its axis is called the vertex.

The vertex is midway between the focus and the directrix.

(112)



This is a parabola with the axis on the x -axis and the focus at a point p units to the right of the origin, where $p > 0$. It follows that the directrix is p units to the left of the origin and so has the equation $x = -p$.

From the definition of a parabola, a point $P(x,y)$ will be on the parabola iff $\overline{PF} = \overline{PP'}$, where F is the focus and P' is the horizontal projection of P onto the directrix and so has the coordinates $(-p, y)$.

Using the distance formula, we have

$$\sqrt{(x-p)^2 + y^2} = x+p.$$

$$\text{i.e. } ((x-p)^2 + y^2)^{1/2} = x+p.$$

(113)

Squaring both sides, we obtain

$$(x-p)^2 + y^2 = (x+p)^2$$

$$x^2 - 2px + p^2 + y^2 = x^2 + 2px + p^2$$

$$\text{or } y^2 = 4px.$$

In Summary, if a parabola with vertex at the origin has a horizontal axis, its equation is of the form $y^2 = \pm 4px$, opening to the right if the sign is plus and the left if the sign is minus.

If the axis is vertical, then the equation is of the form $x^2 = \pm 4py$, opening upward if the sign is plus and downward if it is minus.

NB: Standard form of equations of parabolas

$$(y-k)^2 = \pm 4p(x-h), \text{ axis horizontal.}$$

$$(x-h)^2 = \pm 4p(y-k), \text{ axis vertical.}$$

Recall that the number p is the distance from the vertex to the focus or, the distance from the vertex to the directrix.

For the above cases, the vertex is at the point (h, k)

(114)

The line segment through the focus of a parabola, perpendicular to its axis and terminating on the parabola is called the latus rectum of the parabola. The length of the latus rectum must be $4p$.

Example:

1. Find the equation in standard form of each of the following parabolas and sketch their graphs.

(a) Focus $(-3, 2)$, directrix: $x = 1$.

Solution

Since the directrix is vertical, the axis is horizontal. The focus is to the left of directrix $\Rightarrow$ the parabola opens to the left.

$$\text{i.e. } (y-k)^2 = -4p(x-h)$$

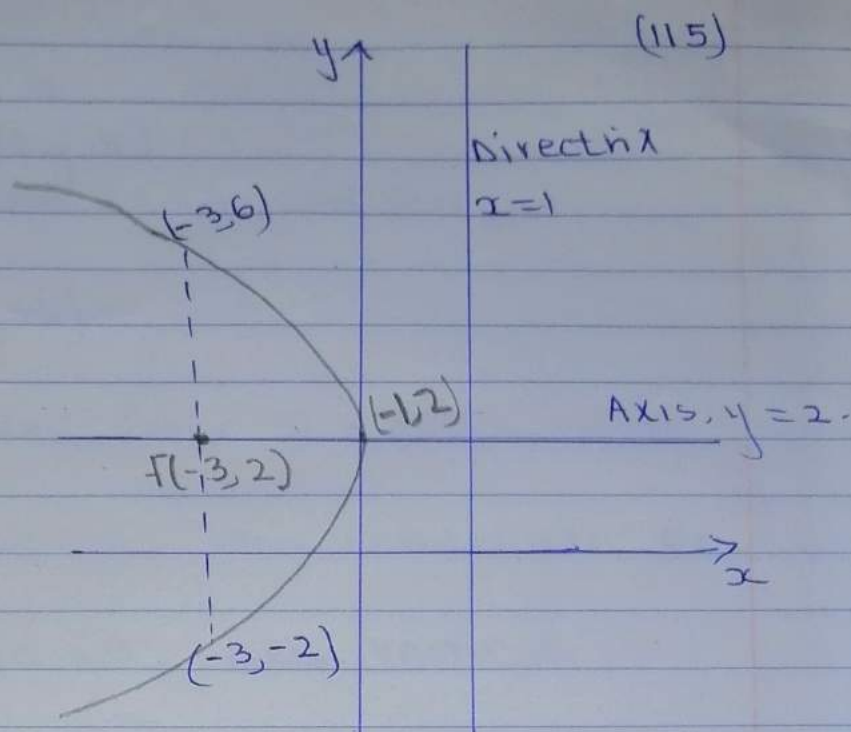
The distance from focus to the directrix is $2p$ and thus

$$2p = 4 \Rightarrow p = 2.$$

The vertex is midway between the focus and directrix $\Rightarrow (h, k) = (-1, 2)$

$\Rightarrow$ The equation now becomes

$$(y-2)^2 = -8(x+1).$$

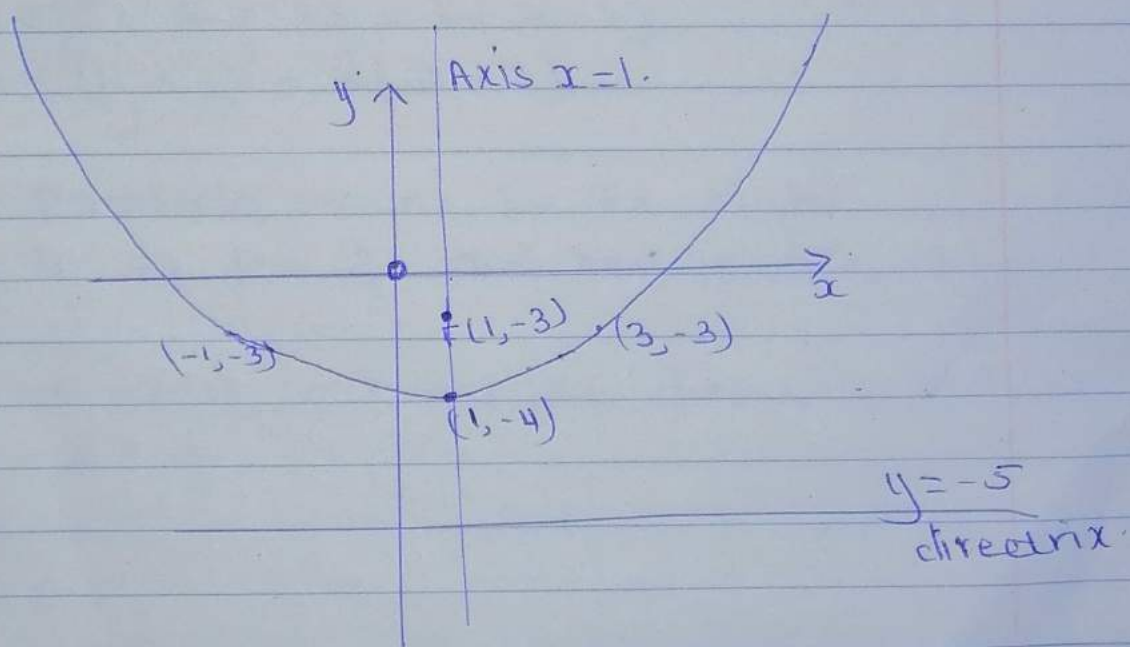


(b) Vertex $(1, -4)$; end points of latus rectum $(-1, -3)$ and $(3, -3)$.

Solution

The length of latus rectum is $4p$
 $\Rightarrow 4p = 4 \Rightarrow p = 1$.

Since vertex is $(1, -4)$, ~~is~~
 the focus is $(1, -3)$



The equation is

$$(x-h)^2 = 4p(y-k)$$

With $h=1$ and $k=-4$, and $p=1 \Rightarrow$

$$(x-1)^2 = 4(y+4).$$

The expanded form of a parabola is

$$y^2 + ax + by + c = 0, \text{ axis horizontal or}$$

$$x^2 + ax + by + c = 0, \text{ axis vertical.}$$

2. Discuss and sketch the graph of each of the following

(a) $y^2 - 6x + 4y + 16 = 0$.

Solution

Completing the square in y

$$y^2 + 4y = 6x - 16$$

$$y^2 + 4y + 4 = 6x - 16 + 4$$

$$(y+2)^2 = 6x - 12 = 6(x-2)$$

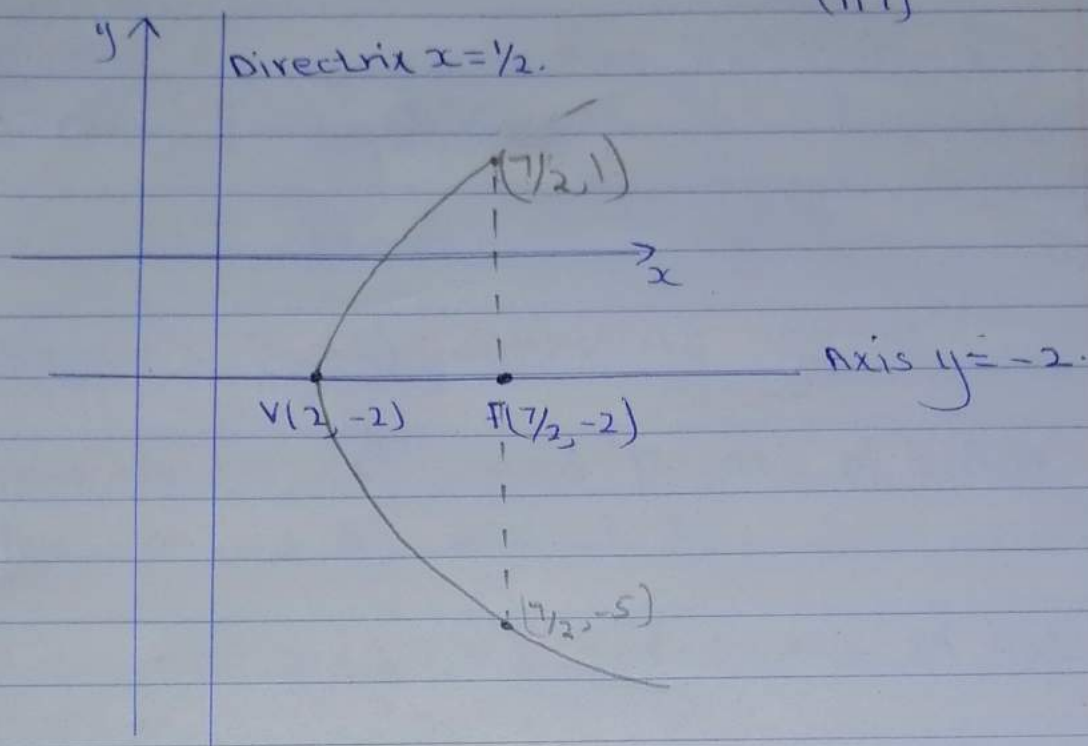
$$\Rightarrow (y+2)^2 = 6(x-2)$$

This parabola opens to the right.

$$4p = 6 \Rightarrow p = \frac{3}{2} \text{ and Vertex} = (2, -2)$$

$\Rightarrow$ Focus is $(\frac{7}{2}, -2)$ and the directrix is the line $x = \frac{1}{2}$

(117)



(b) $3x^2 - 5x + 2y - 4 = 0$

Solution

$$3x^2 - 5x = -2y + 4$$

$$\Rightarrow x^2 - \frac{5}{3}x = -\frac{2}{3}y + \frac{4}{3}$$

$$\left(x - \frac{5}{6}\right)^2 = -\frac{2}{3}y + \frac{4}{3} + \frac{25}{36}$$

$$\left(x - \frac{5}{6}\right)^2 = -\frac{2}{3}\left(y - \frac{13}{24}\right)$$

$\Rightarrow$ Equation of a parabola with Vertex $\left(\frac{5}{6}, \frac{13}{24}\right)$, opening downwards.

$$4p = \frac{2}{3} \Rightarrow p = p = \frac{1}{6}$$

The focus is $\frac{1}{6}$ unit from the vertex, and the latus rectum extends $\frac{1}{3}$ unit to either side.

Exercise:

(118)

1. Find the equation of the parabola described.

(a) Focus $(2, 0)$; directrix $x = -2$

(b) Focus $(0, -3)$; directrix $y = 3$.

(c) Vertex $(1, -2)$; directrix $y = -4$.

(d) Vertex $(2, -3)$; end points of Latus rectum $(-4, 0)$ and $(8, 0)$

(e) Directrix $x + 2 = 0$; end points of Latus rectum $(1, 2)$ and $(1, -4)$.

2. Write the equation in standard form, give the coordinates of the focus, the Latus rectum and the equation of the directrix, sketch the graph.

(a) $y^2 + 8x - 6y + 41 = 0$

(b) $x = y^2 - 3y + 4$

(c) $4y^2 + 4y + 24x - 35 = 0$